
FastAPI Application Server (app.py, 3 285 LOC)
Asynchronous routing • Server-Sent Events (SSE) streaming • GZip compression • CORS allowlist
routes/votes.py
Public voting API
Salted client hash
catalog_export
IVOA Simple Cone
VOTable &
CSV export
routes/deep.py
NEP deep-field
Forced photometry
routes/metrics.py
/api/metrics
Eval metrics & model
Bounded Background Job Engine (jobs.py)
Submitted
Running (1)
Completed
Queue: 1 running, 8 waiting; 128-
bit random IDs; 420 s timeout
Security & Observability (http_guard.py)
• In-Memory Rate Limiter: 30 req/min for votes, 60 req/min for cone search.
• Sanitised Errors: _public_error() strips internal filepaths and URLs.
• Structured Logs (logging_setup.py): JSON logs with request IDs.